

geg. $E[X_i] = 5011$, $D(X_i) = 4000$

1. $n = ?$: $P(X_1 + \dots + X_n > 10^6)$

$$= P\left(\frac{S_n - E(S_n)}{\sqrt{D(S_n)}} > \frac{10^6 - 5011 \cdot n}{\sqrt{4000n}}\right)$$

$S_n := \sum_{i=1}^n X_i$

$$\stackrel{\text{LPT}}{\approx} P\left(N(0,1) > \frac{10^6 - 5011n}{\sqrt{4000n}}\right) = 99\%$$

$$\Rightarrow \frac{10^6 - 5011n}{\sqrt{4000n}} < -2,33$$

$$\Rightarrow \boxed{n \geq 200}$$

2. $n = ?$: $P(\max_{i \in \{1, \dots, n\}} X_i > 10^4) \geq 99\%$

$$\Leftrightarrow P(\max X_i < 10^4) \leq 1\%$$

$$\Leftrightarrow [P(X_i \leq 10^4)]^n \leq 1\%$$

$$\Rightarrow \text{Also } P(X_1 \leq 10^4) =: p, \text{ so}$$

$$\boxed{n \geq \frac{\ln(\frac{1}{100})}{\ln(p)}}$$

Ако искаме конкретна стойност за p , трябва да изберем конкретна разпрег. за X_1 :

Ако $X_1 \sim N(5011, 4000)$

$$P(X_1 \leq 10^4) = P(N(0,1) \leq \frac{10^4 - 5011}{\sqrt{4000}})$$

$$\approx P(N(0,1) \leq 78,88)$$

$$= \Phi(78,88)^n$$

Визуални решения, които стават го израз от сходен тип се очакват с max възможни точки.

бонус:

Ако искаме да намерим явна стойност за n , такава че $\Phi(78,88)^n \leq 1\%$

Макар че $\Phi(78,88)$ е

много близо до 1, повдигнато на

достатъчно висока степен, $\Phi(78,88)^n$ ще стане близо до 0.

Стандартна оценка за $P(U(0,1) > x)$

$$= 1 - \Phi(x) \quad \text{ca}$$

$$\left(\frac{1}{x} - \frac{1}{x^3}\right) \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \leq 1 - \Phi(x) \leq \frac{1}{x} e^{-\frac{x^2}{2}} \cdot \frac{1}{\sqrt{2\pi}}$$

Доказ: $1 - \Phi(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$

$$= \frac{1}{\sqrt{2\pi}} \left(- \int_x^\infty \frac{1}{t} d e^{-\frac{t^2}{2}} \right)$$

по частям $= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-\frac{x^2}{2}} + \int_x^\infty e^{-\frac{t^2}{2}} d \frac{1}{t} \right)$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-\frac{x^2}{2}} - \int_x^\infty \frac{1}{t^2} e^{-\frac{t^2}{2}} dt \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-\frac{x^2}{2}} + \int_x^\infty \frac{1}{t^3} d e^{-\frac{t^2}{2}} \right)$$

по частям $= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-\frac{x^2}{2}} - \frac{1}{x^3} e^{-\frac{x^2}{2}} - \int_x^\infty e^{-\frac{t^2}{2}} d \frac{1}{t^3} \right)$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} - \frac{1}{x^3} \right) e^{-\frac{x^2}{2}} - \frac{1}{\sqrt{2\pi}} \cdot 3 \int_x^\infty \frac{1}{t^4} e^{-\frac{t^2}{2}} dt$$

$$\geq \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} - \frac{1}{x^3} \right) e^{-\frac{x^2}{2}}$$

$$\Rightarrow \Phi(78,88) \leq \left(1 - \left(\frac{1}{x} - \frac{1}{x^3} \right) \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right)$$

за $x = 78,88$

$\Rightarrow$ достаточно для каждого n : за $x = 78,88$

$$\left(1 - \left(\frac{1}{x} - \frac{1}{x^3} \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right)^n \leq 1\%$$

$$\Leftrightarrow n \cdot \ln \left(1 - \left(\frac{1}{x} - \frac{1}{x^3} \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right) \leq 1\%$$

т.к.
 $\ln(1-x) \sim -x$
 $\ln(x) \sim x$

$$n \left(- \left(\frac{1}{x} - \frac{1}{x^3} \right) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right) \leq \ln \left(\frac{1}{100} \right)$$

$$n \geq \ln(100) \cdot e^{\frac{x^2}{2}} \sqrt{2\pi} \cdot \frac{x^3}{x^2 - 1}$$

$$\Rightarrow n \geq 4,16 \cdot 10^{1354} !$$

заг.2 Дам заг.3 изпит СЕМ 2020/21
жыару

заг.3 $X \sim \text{Exp}(\lambda^{-1})$, $Y \sim \text{Exp}(\beta^{-1})$

1.) Төрөөм c : $P(|\lambda - \bar{X}| > c) \leq 1\%$

От $\mathbb{C}^n$, за n

$$\frac{X - \lambda}{\lambda \sqrt{n}} \approx Z \sim N(0, 1) \text{ . Зароо}$$

Төрөөм c : $P(|Z| > \frac{c}{\lambda \sqrt{n}}) \leq 1\%$

$$\Leftrightarrow P(Z > \frac{c}{\lambda \sqrt{n}}) \leq 0.5\%$$

$$\Rightarrow \frac{c}{\lambda \sqrt{n}} \approx 2.57 \Rightarrow c \approx 2.57 \cdot \lambda \sqrt{n}$$

$$2) Z_i := \frac{X_i}{\sqrt{Y_i}} \text{ . Ага } E[Z_i] < \infty, \text{ or}$$

$$\text{3) } \frac{Z_1 + \dots + Z_n}{n} \xrightarrow[n \rightarrow \infty]{\text{n.c.}} E[Z_1]$$

(for values) $P\left(\sqrt{\frac{X}{Y}} > t\right) = P(X > Y t^2)$

$$= \int_0^{\infty} P(X > y t^2) \cdot f_Y(y) dy$$

$P(\text{Exp}(\lambda) > t) = e^{-\lambda t}$

$$= \int_0^{\infty} e^{-\frac{y t^2}{\alpha}} \cdot \frac{1}{\beta} e^{-\frac{1}{\beta} y} dy$$

$$= \left(\frac{t^2}{\alpha} + \frac{1}{\beta}\right)^{-1} \int_0^{\infty} e^{-y \left(\frac{t^2}{\alpha} + \frac{1}{\beta}\right)} dy \left(\frac{t^2}{\alpha} + \frac{1}{\beta}\right)$$

$\int_0^{\infty} e^{-x} dx = 1$

$$= \frac{\alpha \beta}{\beta t^2 + \alpha} \cdot \frac{1}{\beta} = \frac{\alpha}{\beta t^2 + \alpha}$$

$$\Rightarrow E\left[\sqrt{\frac{X}{Y}}\right] = \int_0^{\infty} P\left(\sqrt{\frac{X}{Y}} > t\right) dt$$

$$= \int_0^{\infty} \frac{\alpha}{\beta t^2 + \alpha} dt$$

$$= \int_0^{\infty} \frac{1}{\frac{\beta}{\alpha} t^2 + 1} dt$$

$$y = \sqrt{\frac{\beta}{\alpha}} t \Rightarrow \sqrt{\frac{\alpha}{\beta}} \int_0^{\infty} \frac{1}{y^2 + 1} dy = \sqrt{\frac{\alpha}{\beta}} \cdot \frac{\pi}{2}$$

$\neq \sqrt{\frac{\alpha \pi}{\beta}} = \sqrt{\frac{\alpha}{\beta}}$

2^{te} nærun: $f_{X,Y}(x,y) \stackrel{\text{teq.}}{=} f_X(x) f_Y(y)$

$$= \frac{1}{\alpha} e^{-\frac{1}{\alpha}x} \cdot \frac{1}{\beta} e^{-\frac{1}{\beta}y}$$

$$\Rightarrow H\left[\sqrt{\frac{X}{Y}}\right] = \int_0^\infty \int_0^\infty \sqrt{\frac{x}{y}} \cdot \frac{1}{\alpha} e^{-\frac{1}{\alpha}x} \cdot \frac{1}{\beta} e^{-\frac{1}{\beta}y} dx dy$$

$$= \int_0^\infty \sqrt{x} \frac{1}{\alpha} e^{-\frac{1}{\alpha}x} dx \cdot \int_0^\infty \sqrt{\frac{1}{y}} \cdot \frac{1}{\beta} e^{-\frac{1}{\beta}y} dy$$

$$x' = \frac{x}{\alpha}; y' = \frac{y}{\beta}$$

$$\int_0^\infty \sqrt{x'} \cdot \frac{1}{\sqrt{x'}} e^{-x'} d(x') \quad \checkmark$$

$$\cdot \int_0^\infty \sqrt{\frac{1}{y'}} \cdot \frac{1}{\sqrt{y'}} \cdot \frac{1}{\beta} \cdot e^{-y'} d(y') \quad \checkmark$$

$$P(z) = \int_0^\infty \frac{1}{x} e^{-x} dx$$

$$P(z+1) = zP(z) = \sqrt{\frac{\alpha}{\beta}} P\left(\frac{3}{2}\right) \cdot P\left(\frac{1}{2}\right)$$

$$= \sqrt{\frac{\alpha}{\beta}} \cdot P\left(\frac{1}{2}\right)^2 \cdot \frac{1}{2}$$

$$= \sqrt{\frac{\alpha}{\beta}} \cdot \frac{\pi}{2}$$

3-tes naxun: $\left| \begin{array}{l} z = \sqrt{\frac{x}{y}} \\ w = \sqrt{y} \end{array} \right. \xRightarrow{2^{wsq}} \left\{ \begin{array}{l} x = z^2 \cdot w^2 \\ y = w^2 \end{array} \right.$

$$|J| = \begin{vmatrix} 2zw^2 & 2z^2w \\ 0 & 2w \end{vmatrix} = 4zw^3$$

$$\Rightarrow f_{zw}(z, w) = f_{xy}(z^2w^2, w^2) \cdot 4zw^3$$

$$= \frac{1}{\alpha\beta} e^{-\frac{1}{2}z^2w^2 - \frac{1}{\beta}w^2} \cdot 4zw^3$$

$$\rightarrow f_z(z) = \int_0^\infty \frac{1}{\alpha\beta} e^{-w^2(\frac{1}{\beta} + \frac{z^2}{2})} \cdot 4zw^3 dw$$

$$= \frac{4z}{\alpha\beta} \cdot \frac{1}{2} \int_0^\infty e^{-w^2(\frac{1}{\beta} + \frac{z^2}{2})} w^2 dw$$

$$\begin{aligned} w^2(\frac{1}{\beta} + \frac{z^2}{2}) &= y & \frac{2z}{\alpha\beta} \cdot \left(\frac{1}{\beta} + \frac{z^2}{2}\right)^{-2} \cdot \int_0^\infty e^{-y} y dy \\ & & \underbrace{\int_0^\infty e^{-y} y dy}_{=1} \\ & & = E[Exp(1)] \end{aligned}$$

$$= \frac{2z}{\alpha\beta} \cdot \frac{(\alpha\beta)^2}{(\alpha + \beta z^2)^2}$$

$$= 2\alpha\beta \cdot \frac{z}{(\alpha + \beta z^2)^2} = 2 \cdot \frac{\beta}{\alpha} \cdot \frac{z}{(1 + \frac{\beta}{\alpha} z^2)^2}$$

$$\Rightarrow F[z] = \int_0^\infty z \cdot 2 \cdot \frac{\beta}{\alpha} \cdot \frac{z}{(1 + \frac{\beta}{\alpha} z^2)^2} dz$$

$$\begin{aligned} z = \sqrt{\frac{\alpha}{\beta}} y & \Rightarrow \int_0^\infty \sqrt{\frac{\alpha}{\beta}} y \cdot 2 \cdot \frac{\beta}{\alpha} \cdot \sqrt{\frac{\alpha}{\beta}} y \cdot \frac{1}{(1 + y^2)^2} d\sqrt{\frac{\alpha}{\beta}} y \\ & = \int_0^\infty \sqrt{\frac{\alpha}{\beta}} y \cdot 2 \cdot \frac{\beta}{\alpha} \cdot \sqrt{\frac{\alpha}{\beta}} y \cdot \frac{1}{(1 + y^2)^2} d\sqrt{\frac{\alpha}{\beta}} y \end{aligned}$$

$$\begin{aligned} &= 2\sqrt{\frac{\alpha}{\beta}} \int_0^\infty \frac{y^2}{(1 + y^2)^2} dy = 2\sqrt{\frac{\alpha}{\beta}} \cdot \frac{\pi}{4} = \\ & \quad \frac{\pi}{2} \cdot \sqrt{\frac{\alpha}{\beta}}. \end{aligned}$$

зая. 4

Вам зая. 4 от узник СЕМ
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